

-- Cox Regression - Sarcopenia Onset -----

i Definition : fnih
i Covariates : time_dependent
i Dairy : categorical
i Route : mice
i Interaction: none

-- MICE route - pooling across imputed datasets --

Detected 20 imputed datasets.
Fitting .imp = 1 / 20 ...
Events: 34 sarcopenia cases out of 1117 participants

-- Time-dependent covariates dataset (one row per pt x visit) --

dairy_exposure levels: "Q1", "Q2", "Q3", and "Q4"
??? Time-dependent dataset: 1951 intervals | 32 events | 7798.4 person-years
Fitting .imp = 2 / 20 ...
Events: 34 sarcopenia cases out of 1117 participants

-- Time-dependent covariates dataset (one row per pt x visit) --

dairy_exposure levels: "Q1", "Q2", "Q3", and "Q4"
??? Time-dependent dataset: 1951 intervals | 32 events | 7798.4 person-years
Fitting .imp = 3 / 20 ...
Events: 35 sarcopenia cases out of 1117 participants

-- Time-dependent covariates dataset (one row per pt x visit) --

-- Time-dependent covariates dataset (one row per pt x visit) --

dairy_exposure levels: "Q1", "Q2", "Q3", and "Q4"
??? Time-dependent dataset: 1950 intervals | 33 events | 7795.8 person-years
Fitting .imp = 6 / 20 ...
Events: 35 sarcopenia cases out of 1117 participants

-- Time-dependent covariates dataset (one row per pt x visit) --

dairy_exposure levels: "Q1", "Q2", "Q3", and "Q4"
??? Time-dependent dataset: 1950 intervals | 33 events | 7795.8 person-years
Fitting .imp = 7 / 20 ...
Events: 36 sarcopenia cases out of 1117 participants

-- Time-dependent covariates dataset (one row per pt x visit) --

dairy_exposure levels: "Q1", "Q2", "Q3", and "Q4"
??? Time-dependent dataset: 1950 intervals | 34 events | 7795.8 person-years
Fitting .imp = 8 / 20 ...
Events: 34 sarcopenia cases out of 1117 participants

-- Time-dependent covariates dataset (one row per pt x visit) --

dairy_exposure levels: "Q1", "Q2", "Q3", and "Q4"
??? Time-dependent dataset: 1951 intervals | 32 events | 7798.4 person-years
Fitting .imp = 9 / 20 ...
Events: 36 sarcopenia cases out of 1117 participants

dairy_exposure levels: "Q1", "Q2", "Q3", and "Q4"
??? Time-dependent dataset: 1951 intervals | 33 events | 7798.4 person-years
Fitting .imp = 11 / 20 ...
Events: 35 sarcopenia cases out of 1117 participants

-- Time-dependent covariates dataset (one row per pt x visit) --

dairy_exposure levels: "Q1", "Q2", "Q3", and "Q4"
??? Time-dependent dataset: 1951 intervals | 33 events | 7798.4 person-years
Fitting .imp = 12 / 20 ...
Events: 35 sarcopenia cases out of 1117 participants

-- Time-dependent covariates dataset (one row per pt x visit) --

dairy_exposure levels: "Q1", "Q2", "Q3", and "Q4"
??? Time-dependent dataset: 1950 intervals | 33 events | 7795.8 person-years
Fitting .imp = 13 / 20 ...
Events: 34 sarcopenia cases out of 1117 participants

-- Time-dependent covariates dataset (one row per pt x visit) --

dairy_exposure levels: "Q1", "Q2", "Q3", and "Q4"
??? Time-dependent dataset: 1951 intervals | 32 events | 7798.4 person-years
Fitting .imp = 14 / 20 ...
Events: 34 sarcopenia cases out of 1117 participants

-- Time-dependent covariates dataset (one row per pt x visit) --

dairy_exposure levels: "Q1", "Q2", "Q3", and "Q4"
??? Time-dependent dataset: 1951 intervals | 32 events | 7798.4 person-years

-- Time-dependent covariates dataset (one row per pt x visit) --

dairy_exposure levels: "Q1", "Q2", "Q3", and "Q4"
??? Time-dependent dataset: 1951 intervals | 32 events | 7798.4 person-years
Fitting .imp = 17 / 20 ...
Events: 34 sarcopenia cases out of 1117 participants

-- Time-dependent covariates dataset (one row per pt x visit) --

dairy_exposure levels: "Q1", "Q2", "Q3", and "Q4"
??? Time-dependent dataset: 1951 intervals | 32 events | 7798.4 person-years
Fitting .imp = 18 / 20 ...
Events: 35 sarcopenia cases out of 1117 participants

-- Time-dependent covariates dataset (one row per pt x visit) --

dairy_exposure levels: "Q1", "Q2", "Q3", and "Q4"
??? Time-dependent dataset: 1950 intervals | 33 events | 7795.8 person-years
Fitting .imp = 19 / 20 ...
Events: 34 sarcopenia cases out of 1117 participants

-- Time-dependent covariates dataset (one row per pt x visit) --

dairy_exposure levels: "Q1", "Q2", "Q3", and "Q4"
??? Time-dependent dataset: 1951 intervals | 32 events | 7798.4 person-years
Fitting .imp = 20 / 20 ...
Events: 34 sarcopenia cases out of 1117 participants

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-- Events Per Variable (EPV) [mice (all imputations)]
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i Events : 32
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i Coefs : 10
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i EPV : 3.2
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-- C-index (Harrell's concordance) [mice (averaged)]
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i C-index : 0.804 (95% CI: 0.745-0.863)
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i SE : 0.0299
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??? C-index >= 0.70 - good discrimination.
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-- Global model tests [mice (averaged)]
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i Likelihood ratio test :  $\chi^2(10) = 31.18$ ,  $p = 0.000561$  <- preferred
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i Wald test :  $\chi^2(10) = 24.69$ ,  $p = 0.00608$ 
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i Score (log-rank) test :  $\chi^2(10) = 31.9$ ,  $p = 0.000425$ 
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i Total person-years (mean across imputations): 7797.6
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-- Assumption checks [mice_imp1] --
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-- 1. Proportional Hazards - cox.zph() | pass: global  $p > 0.05$ 
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-- 2. Linearity - martingale (null model) vs continuous predictors | pass: smooth ??? linear
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```
'geom_smooth()' using formula = 'y ~ x'
```

```
'geom_smooth()' using formula = 'y ~ x'
```

```
Linearity plots saved to
```

```
03_outputs/Cox/mice_fnih_time_dependent_categorical_27072026_1811
```

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-- 3. Outliers - deviance & martingale residuals | pass:  $|dev| \leq 3$ 
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-- 7. Log-log survival plot | pass: parallel lines across log(time)

Log-log plot saved to

03_outputs/Cox/mice_fnih_time_dependent_categorical_27072026_1811

-- 8. Cox-predicted vs KM - calibration by dairy quartile

Cox vs KM calibration plot saved to

03_outputs/Cox/mice_fnih_time_dependent_categorical_27072026_1811

-- Kaplan-Meier plot by dairy category [mice_imp1]

i KM (td): baseline dairy quartile x final event status (32 events in 1117
pts).

Ignoring unknown labels:

o colour : "Dairy quartile"

i Log-rank p-value: 0.904

KM plot saved to

03_outputs/Cox/mice_fnih_time_dependent_categorical_27072026_1811

-- Adjusted scenario survival curves (time-dependent) [mice_imp1]

-- Incidence rates by dairy quartile [mice]

quartile	n_pts	n_events	person_years	rate_per_1000py	ci_low	ci_high
1 Q1	268	7.3	1878.00	3.89	1.60	7.89
2 Q2	292	8.0	2050.00	3.90	1.68	7.69
3 Q3	273	9.3	1891.12	4.92	2.28	9.24
4 Q4	284	8.0	1978.50	4.04	1.75	7.97

Incidence table saved to

03_outputs/Cox/mice_fnih_time_dependent_categorical_27072026_1811

Model results saved to

03_outputs/Cox/mice_fnih_time_dependent_categorical_27072026_1811/mice_model_results.csv